That in simple words means that the world will be generating more than 463 exabytes of data every day! Please note, the world here denotes data created by humans, machines, and the internet.

"90% of the data generated, by the end of this decade, will be by machines. Machines themselves are creating a jump in the amount of data, then humans are also driving this growth. As the amount of data generation grows, so will the need for computing. This will lead to a tremendous demand for semiconductors. I think this has been the concern of a lot of policy makers. The rate at which the demand of semiconductor is growing, it can play havoc at how national economies function," explains Ashwini.

Our country relies 100% on imports for meeting its semiconductor needs. The value of semiconductors that India will be importing by 2025 is forecast to be around $100 billion. China, which was till recently referred to as the world's electronics factory, imported around $378 billion worth of semiconductors during the year 2020. China currently holds 7.6% share of semiconductors sold in the world. This is another fact that makes it critical for India to establish a semiconductor manufacturing ecosystem as soon as possible.

The global semiconductor packaging market, as per Allied Market Research, is expected to reach $60.44 billion by 2030 from $27.1 billion in 2020, growing at a CAGR of 9.1% from 2021 to 2030.

"The opportunities are immense. All the stakeholders belonging to India's electronics sector agree that this industry would hit the size of $300 billion by the year 2025. The market for semiconductors by 2030 will go to probably $1.2 trillion. India consumes about five to six percent of the total semiconductors consumed in the world. This number is going to increase, keeping in mind that we represent 18% of the world's population," says Amrit Manwani, MD, Sahasra Group.

The segments which are driving and will continue to drive the growth

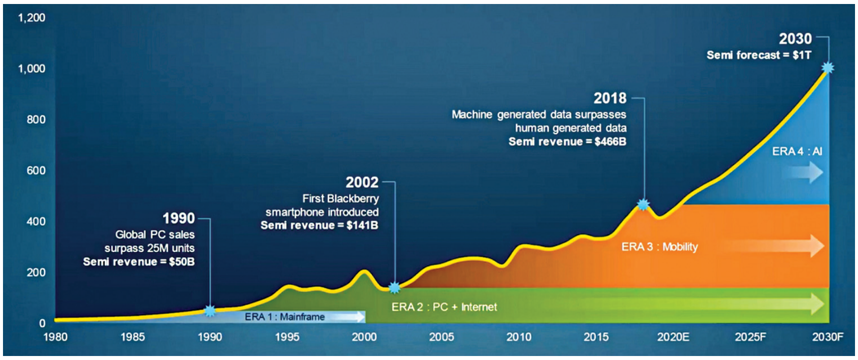


*Fig. 2: Semiconductor industry revenue ($ billion)*

of not just the semiconductors industry but the overall components industry involve mobile phones, IT hardware, LED lighting, power electronic devices, automotive, telecom, consumer electronics, healthcare devices, and defense electronics.

## OSAT before fab

Experts are divided on the best steps towards establishing a semiconductor manufacturing ecosystem in the country. Some opine that the perfect route goes through establishing fabs as the first step, some say OSATs/ATMPs should come first, while others advise on strengthening the semiconductor design ecosystem. Frankly, there are countries where the design ecosystem is stronger than fabs or OSATs, and vice versa as well.

"It will take time for the semiconductor fabs to come into the country. Before we do that downstream, we must have an ecosystem which explains what we are going to do with the wafers. Are we just going to supply these overseas? If this happens then another question will pop up in the form of why we set up the fabs in the first place if we were not able to package them here. It's very important that we have a packaging base established in the country," shares Manwani.

Now, while it's not a rule, setting up an OSAT/ATMP can be achieved at least a year faster than setting up a fabrication plant. It's also a fact that establishing an ATMP requires less investment than a fab, and also not as highly skilled professionals as a fab would require. For example, Sahasra is investing close to ₹1.5 billion in building an ATMP unit, while the Tata's are rumored to be putting in more than ₹3.5 billion in the same. Investing in establishing a fab, in comparison, will require at least a $1 billion!

There are more than 150 companies operating in the memory packaging business in the world, though none in India, and we consume 10 to 12% of the memory chips. But then if OSATs are the right step towards enabling a semiconductor manufacturing ecosystem, why has India not been able to cash-in on this opportunity in the modern times?

"The lead time for equipment required for semiconductor packaging has gone high because of the massive semiconductor shortage worldwide. The dealers are now quoting delivery time of more than 50 days for the equipment deliveries. The prices have increased by about 15 to 20 percent. Process stability as well as keeping up with the new technology are a few more challenges. Yield is another big challenge," explains Manwani.

"One of the reasons that we are discussing semiconductors and India is because we understand that it is a big opportunity for the nation as well as for Indian organisations. It is a huge growth area," says Raja Manickam, CEO, Tata OSAT.

## Breaking myths

The semiconductor industry, probably the most aggressive in terms of growth, is also haunted by a number